

Countdown **TIMER**

FAYAZ HASSAN

Sometimes there is a need to switch on or off electrical appliances or gadgets after a predetermined time. The circuit pre-

sented here can switch on/off any electrical appliance (load) through a relay switch after the specified time set in minutes or seconds.

The timer can be easily set with four switches (S3 through S6) to select minutes/seconds, hundredths, tenths and units. Two more switches

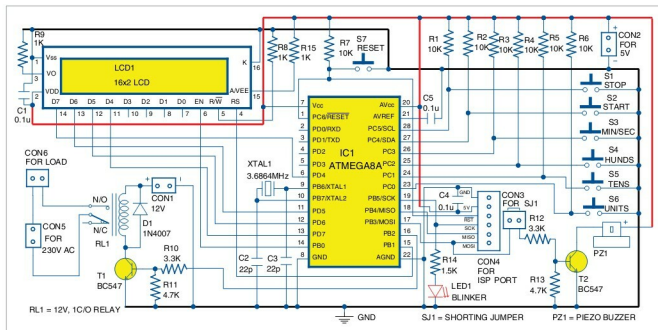


Fig. 1: Circuit diagram of countdown timer

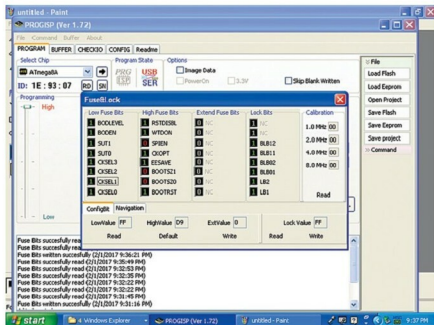


Fig. 2: Screenshot of the fuse bits settings: '1' means 'unprogrammed' and '0' means 'programmed'

are used to start (S2) and stop (S1) the timer.

Circuit and working

Circuit diagram of the countdown timer is shown in Fig. 1. ATmega8A (IC1) from Atmel is at the heart of the circuit. Associated components include a 16×2 liquid crystal display (LCD1), two BC547 transistors (T1 and T2), a piezobuzzer (PZ1), a 12V single-changeover (1C/O) relay and a few other components.

The system works in three modes: started, stopped and setting. Started means the countdown is under process and settings are not allowed. Stopped means the countdown has stopped and the user can change settings to start the countdown timer afresh. In setting mode,